Question~ What is the difference between synchronous and asynchronous programming?
Answer~ Synchronous programming means tasks are executed one after another. Each task waits for the previous task to finish before moving to the next one. This makes the flow simple and predictable, but it can be slow if one task takes a long time, like a network request.
Asynchronous programming allows tasks to run without waiting for previous tasks to complete. For example, if a program makes an API call, it doesn't block the rest of the code while waiting for the response. Instead, it can continue executing other tasks and handle the result later using callbacks, promises, or async/await.
Asynchronous programming improves performance and responsiveness, especially in applications like web servers or user interfaces. However, it can make the code harder to understand and debug due to callbacks and concurrency issues.

Question= Explain what indexing is in a database and why it is used.

Answer= Indexing in a database is a technique used to improve the speed of data retrieval. It works similarly to an index in a book — instead of scanning the entire table to find specific data, the database uses the index to quickly locate the required rows.
An index is usually created on one or more columns of a table. When a query searches for data using those columns, the database can use the index to reduce the number of rows it needs to scan. This significantly improves SELECT query performance, especially for large tables.
However, indexes also have some drawbacks. They take extra storage space and can slow down INSERT, UPDATE, and DELETE operations because the index must also be updated whenever the data changes.

Common types of indexes include primary index and secondary index.